Example 6.38

Find the vector and Cartesian form of the equations of a plane which is at a distance of 12 units from the origin and perpendicular to $6\hat{i}+2\hat{j}-3\hat{k}$.

Solution

Let $\vec{d}=6\hat{i}+2\hat{j}-3\hat{k}$ and $p=12$.

If $\hat{d}$ is the unit normal vector in the direction of the vector $6\hat{i}+2\hat{j}-3\hat{k}$,

then $\hat{d}=\dfrac{\vec{d}}{|\vec{d}|}=\dfrac{1}{7}(6\hat{i}+2\hat{j}-3\hat{k})$.

If $\vec{r}$ is the position vector of an arbitrary point (x,y,z) on the plane, then using $\vec{r}\cdot\hat{d}=p$, the vector equation of the plane in normal form is $\vec{r}\cdot\dfrac{1}{7}(6\hat{i}+2\hat{j}-3\hat{k})=12$.

Substituting $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$ in the above equation, we get $(x\hat{i}+y\hat{j}+z\hat{k})\cdot\dfrac{1}{7}(6\hat{i}+2\hat{j}-3\hat{k})=12$.

Applying dot product in the above equation and simplifying, we get $6x+2y-3z=84$, which is the Cartesian equation of the required plane.

Example 6.39

If the Cartesian equation of a plane is $3x-4y+3z=-8$, find the vector equation of the plane in the standard form.

Solution

If $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$ is the position vector of an arbitrary point (x,y,z) on the plane, then the given equation can be written as $(x\hat{i}+y\hat{j}+z\hat{k})\cdot(3\hat{i}-4\hat{j}+3\hat{k})=-8$ or $(x\hat{i}+y\hat{j}+z\hat{k})\cdot(-3\hat{i}+4\hat{j}-3\hat{k})=8$. That is, $\vec{r}\cdot(-3\hat{i}+4\hat{j}-3\hat{k})=8$ which is the vector equation of the given plane in standard form.

Example 6.40

Find the direction cosines of the normal to the plane and length of the perpendicular from the origin to the plane $\vec{r}\cdot(3\hat{i}-4\hat{j}+12\hat{k})=5$.

Solution

Let $\vec{d}=3\hat{i}-4\hat{j}+12\hat{k}$ and $q=5$.

If $\hat{d}$ is the unit vector in the direction of the vector $3\hat{i}-4\hat{j}+12\hat{k}$, then $\hat{d}=\dfrac{1}{13}(3\hat{i}-4\hat{j}+12\hat{k})$.

Now, dividing the given equation by 13, we get

$$\vec{r}\cdot\left(\frac{3}{13}\hat{i}-\frac{4}{13}\hat{j}+\frac{12}{13}\hat{k}\right)=\frac{5}{13}$$

which is the equation of the plane in the normal form $\vec{r}\cdot\hat{d}=p$.

From this equation, we infer that $\hat{d}=\dfrac{1}{13}(3\hat{i}-4\hat{j}+12\hat{k})$ is a unit vector normal to the plane from the origin. Therefore, the direction cosines of $\hat{d}$ are $\dfrac{3}{13},\dfrac{-4}{13},\dfrac{12}{13}$ and the length of the perpendicular from the origin to the plane is $\dfrac{5}{13}$.

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